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Analysis of per- and polyfluoroalkyl substances (PFAS) in raw materials intended for the production of paper-based food contact materials – evaluating LC-MS/MS versus total fluorine and extractable organic fluorine

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ABSTRACT

Per- and polyfluoroalkyl substances (PFAS) analysis has become crucial due to their presence in the environment, their persistence and potential health risks. These compounds are commonly used in food contact materials (FCM) as a coating to provide water and grease-repellent properties. One of the pathways for PFAS to enter the human body is either through direct consumption of contaminated food or indirectly through migration from FCM into food. The purpose of this study was to investigate where the initial contamination of paper FCM occurs. We analysed paper material consisting of fresh fibre and secondary materials, intended to produce food packaging for the presence of PFAS. The samples were extracted and analysed for 23 different PFAS substances using the targeted approach with LC tandem mass spectrometry (LC-MS/MS). This analytical technique detects specific, easily ionisable PFAS with high sensitivity. However, one drawback of this approach is that it allows the identification of less than 1% of the PFAS known today. For this reason, we used combustion ion chromatography (CIC) to determine the content of extractable organic fluorine compounds (EOF) and compare it to the total fluorine content. The targeted analysis using LC-MS/MS measured an average sum concentration of PFAS of 0.17 ng g^{-1} sample. Our research shows that the primary PFAS contamination happens during the recycling process since all of the samples in which the targeted PFAS were measured belonged to the secondary material. The most frequently detected analytes were PFOA and PFOS, detected in 90% and 62% of the samples, respectively, followed by PFBS (in 29% of the samples). CIC showed that measured PFAS via LC-MS/MS amount to an average of 2.7×10^{-4}% of total fluorine content, whereas the EOF was under the LOD in all of the measured samples. This result highlights the complexity of the accurate determination of PFAS compounds, displaying what kind of information the chosen methods provide.

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Introduction

Per- and polyfluoroalkyl substances (PFAS) have been recognized as an emerging topic of concern for several decades. PFAS are a group of human-made chemicals that have been widely used for the production of various industrial and consumer products (Bečanová et al. 2016). Due to their unique water- and grease-resistant properties, PFAS are valuable substances in applications like firefighting foams, non-stick cookware, waterproof fabrics, food packaging materials, and more (Glüge et al. 2020). The bond between the carbon and the fluorine is extremely rare in nature and PFAS are exclusively man-made. The strong bond energy between C and F gives PFAS exceptional stability (Key et al. 1997). However, because of their stability, most PFAS break down very slowly over time, although some may undergo partial degradation. These persistent PFAS, together with their breakdown products, pose a significant risk because once they are released into the environment, they can potentially contaminate the food chain for decades. According to a chemical database maintained by the U.S. EPA, currently, there are nearly 15,000 known structures of PFAS

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(Environmental Protection Agency 2021). Additionally, for many substances there is no information about their structure, partially due to the many precursors and unknown intermediate products, and partially because they are patented by the companies that produce them (Göckener et al. 2023). Therefore, comprehensive identification and quantification of PFAS presents considerable analytical challenge.

PFAS have been detected in many matrices, including water, soil, air and biota. Since the 1940s these substances have been used in the production of various food packaging materials due to their properties, which help prevent foods from sticking to packaging and packaging from soaking up the contents (Bokkers et al. 2019). One of the major pathways of human exposure to PFAS is food consumption (Domingo and Nadal 2019). Potential sources of PFAS in food items include contaminated water used for irrigation, contaminated soil, and the use of PFAS-containing biosolids as fertilisers (Ogunbiyi et al. 2023). Over the past decade, several studies have shown that food packaging materials can contain PFAS (Moreta and Tena 2013; Bloom and Hanssen 2015; Zabaleta et al. 2020). Furthermore, it has also been shown that these chemicals can migrate from food packaging into the food (Trier et al. 2018; Ramírez Carnero et al. 2021). Epidemiological studies have linked PFAS exposure to a range of adverse health effects. These effects can vary depending on the specific type of PFAS and the level of exposure and may include different types of cancer, a weakened immune system, liver disease, and reproductive and developmental effects (Fenton et al. 2021). However, for most PFAS there are no toxicological data available. Some of the PFAS concentrations reported in food, including drinking water, and food packaging are summarized in Table 1.

In response to concerns surrounding PFAS in food packaging materials and their potential to migrate to foodstuff, twelve states in the U.S. have issued bans on the use of these substances for the production of FCM. In the European Union, regulations regarding PFAS in FCM fall under the framework of Regulation (EC) No 1935/2004. This regulation establishes the general principles and requirements for materials and articles intended to come into contact with food. It requires that food contact materials do not transfer their constituents to food in quantities that could endanger human health, change the composition of food, or deteriorate its organoleptic characteristics under normal and foreseeable conditions of use. In 2023, five Member States submitted a proposal to the European Chemical Agency (ECHA) to ban the production, placing on the market and use of PFAS in non-essential applications. In June 2020, the Danish Ministry of Environment and Food issued a ban on use of fluorinated substances in paper and board FCM without a functional barrier to prevent migration of the substances to food (Danish Veterinary and Food Administration 2020). The European Food Safety Authority (EFSA) released a recommendation for the limit of the combined exposure to four PFAS (perfluorooctanoic acid (PFOA), perfluorononanoic acid (PFNA), perfluorooctanesulfonic acid (PFOS) and perfluorohexanesulfonic acid (PFHxS)) in food to 4.4 ng kg^{-1} body weight per week (European Food Safety Authority 2020). In 2022, the European Union Reference Laboratory (EURL) for halogenated persistent organic pollutants (POPs) in Feed and Food published a Guidance Document on Analytical Parameters for the Determination of PFAS in Food and Feed (EURL for halogenated POPs in Feed and Food 2022). This document specifies targeted limits of quantification for PFOS, PFOA, PFNA and PFHxS in different food matrices, ranging from 0.001 to 0.02 µg kg^{-1} (*w/w*). To the best of our knowledge, there have been no published methods that fulfil this criterium.

Several analytical methods have been developed for the analysis of PFAS (Al Amin et al. 2020). The most sensitive and selective detection is achieved with LC-MS/MS. However, this technique so far has been reported to monitor only up to 73 different PFAS, limited by the availability of standards and the ionizability of PFAS (Aro et al. 2021). Some non-selective analytical methods can provide more insight into how many PFAS are not detected with LC-MS/MS. For instance, particle-induced gamma-ray emission (PIGE) spectrometry is used for fluorine detection in various matrices (Ritter et al. 2017). However, this technique is non-selective and measures both organic and inorganic fluorine.

Table 1. Overview of levels of PFAS reported in water, food and food packaging materials, obtained via targeted analysis.

Matrices	Source	Concentration	Most detected species	References
Water	Drinking water	<20 ng L^{-1}	PFBS, PFHxS, PFOS, PFHxA, PFOA, PFHpA GenX	(Appleman et al. 2014) (Crone et al. 2019) (Domingo and Nadal 2019)
Food	Meat products	<0.35 ng g^{-1}	PFOS, PFOA, PFDoDA	(Gebbink et al. 2015)
	Wild boar organs	<500 ng g^{-1}	PFNA, PFOA, PFOS	(Schröder et al. 2024)
	Fish	<60 ng g^{-1}	PFOS, PFDoDA PFDA, PFUnDA	(Berger et al. 2009; Valsecchi et al. 2021)
	Milk and milk products	<10 ng g^{-1}	PFBA, PFOA	(Sznajder-Katarzyńska et al. 2019)
	Different vegetables	<0.1 ng g^{-1}	PFOA, PFOS	(Herzke et al. 2013)
Food packaging materials	Microwave popcorn bags	<500 ng g^{-1}	PFOA, PFHxA, PFBA, PFDA, 8:2diPAP	(Gebbink et al. 2013; Zabaleta et al. 2017, 2020)
	Pizza boxes, muffin cups, coated paper	<40 ng g^{-1}	PFHpA, PFOS, PFDA, FTOH	(Moreta and Tena 2013; Granby and Tesdal Håland 2018; Sapozhnikova et al. 2023)
	French fries, popcorn and sandwich wrappers	0.2–48 ng g^{-1}	PFOA, PFHxS, PFHpA, PFHpS, PFOS, PFOSA, N-EtFOSA	(Nxumalo et al. 2023)
	Fast food wrappers	<90–40 ng g^{-1}	PFDA, PFDoA, PFHxA	(Zafeiraki et al. 2014)

High-resolution mass spectrometry can detect more than 750 PFAS, however, it is prone to false positive and false negative results and exhibits high LOD and LOQ (Charbonnet et al. 2021). Finally, combustion ion chromatography (CIC) is commonly used for total fluorine (TF) and extractable organic fluorine analysis. This method has proven to be reliable and robust for total fluorine analysis in an extract (EOF) and directly in a solid sample as total fluorine. However, it is non-selective and provides no information about structure or molecular weight.

PFAS are not naturally present in fresh paper fibres. However, during paper production, they are added to the pulp or used as a surface treatment (Trier et al. 2018). Certain PFAS are non-intentionally added substances (NIAS) and they are thought to originate either from process water or from raw materials used during the recycling process (Curtzwiler et al. 2021), as well as from unknown uses of PFAS in the recycling equipment (Langberg et al. 2024) Several studies have reported the presence of different PFAS in packaging materials using targeted analysis (Wang et al. 2013; Yuan et al. 2016; Zabaleta et al. 2016). Although there are still no official migration limits established for PFAS in food contact materials, estimating the potential migration to food matrices is crucial for risk assessments.

A recent study has shown that PFAS can be found in material coming from a paper recycling plant (Langberg et al. 2024), which highlights the need to investigate raw materials intended for FCM production. This work aims to give insight into where the initial contamination of paper FCM with PFAS occurs. Within this context, 57 samples were analysed, consisting of fresh fibre pulp, virgin paper fibres, secondary pulp and secondary paper. LC-MS/MS analysis was carried out to quantify 23 different compounds, while CIC was used for the analysis of the total fluorine content and extractable organic fluorine. Linearity, recovery and limits of detection and quantification have been established to test the robustness of the methods. Finally, migration potential from the papers into the food has been calculated under the assumption that all of the detected substances would migrate from the packaging to food.

Materials and methods

Materials

Fifty-seven different paper samples intended for the production of food packaging materials were collected between 2021 and 2023. The samples were obtained from different paper and board manufacturers, taken from various stages of the production process, and consisted of fresh fibre pulp, virgin fibre paper, and secondary paper and pulp. The variation in fibre sources helps to understand at which point of FCM production PFAS are potentially introduced in the product. A subset of 20 samples, selected to represent all

four categories of materials, was analysed for total fluorine content and extractable organic fluorine (EOF) using combustion ion chromatography. The complete set of samples was subjected to comprehensive analysis for 23 PFAS compounds using LC-MS/MS. The list of the targeted analytes, as well as the labelled standards used can be found in Supplementary material (Table S1).

Extraction

The samples were extracted with methanol (LC-MS grade, Merck, Germany) using accelerated solvent extraction (ASE) (Thermo Fisher Scientific, Massachusetts, USA). Samples are cut into pieces of approximately 0.5 × 0.5 cm and 2 g are weighed in and placed in ASE cells. The samples are then spiked with 100 µL of 10 ng mL^{-1} isotopically-labelled internal standard (ISTD) mixture in methanol, yielding the total amount of individual internal standard of 1 ng. For EOF measurements, the same sample size was extracted with methanol using ASE, without the addition of the internal standard. Afterwards, the cells are closed tightly and extracted with methanol at 90 °C with three extraction cycles.

After the extraction is completed, the extracts are collected and transferred into evaporation vials and evaporated at 45 °C under a gentle stream of nitrogen until dryness, using the Biotage TurboVap II (Biotage, Uppsala, Sweden) evaporation system. The extracted samples are then reconstituted with 5 ml methanol: water (50:50, *v/v*), and an aliquot is filtered through a syringe filter (Chromafil Xtra, Macherey-Nagel™, Germany) with 0.22 µm regenerated cellulose membrane into polypropylene autosampler vials (Shimadzu, Japan). For EOF measurements, extracts were evaporated to 1 ml of methanol, and filtered. Each sample was analysed in duplicate.

For total fluorine analysis, samples are milled using the cryogenic mixer mill Cryomill (Retsch, Germany) with a stainless-steel ball. Samples are cut into smaller pieces and placed into grinding jars, which are continually cooled with liquid nitrogen. Samples are milled in three cycles of 30 s, with a vibration frequency of 30 s^{-1}. After milling, the samples were kept in glass vials in the refrigerator at 4 °C until analysis.

Quality control

LC-MS/MS

Seven calibration solutions were prepared from a standard solution mixture of native PFAS and isotopically labelled PFAS in methanol (Table S1) (PFAC-24PAR, MPFAC-24ES, respectively, Wellington Laboratories, Canada), where native PFAS concentration ranged between 0.01 and 5 ng mL^{-1} for each analyte. Isotopically labelled PFAS were added to calibration solutions as internal standards in a concentration of 1 ng mL^{-1} for each labelled PFAS. Stock solutions of native PFAS of 100 and 10 ng mL^{-1} were prepared in methanol. These solutions were then used to prepare calibration solutions in methanol: water (50:50, *v/v*). The calibration solutions were measured in triplicates. A solvent blank (methanol: water 50:50, *v/v*) was measured in between each new series of calibration solutions to prevent a possible carry-over. Additionally, a mobile phase blank was also measured before the calibration solutions, to ensure that there are no PFAS present in the device itself, which could potentially cross-contaminate the samples.

A spiking solution was prepared with native PFAS solution (PFAS-24PAR) in a concentration of 10 ng mL^{-1} in methanol, for each PFAS. The internal standard solution was also prepared from isotopically labelled PFAS solution (MPFAC-24ES) in a concentration of 0.01 ng mL^{-1} for each labelled PFAS in methanol. A blank printing paper, which was previously confirmed to contain undetectable target PFAS, was selected as a sample for method validation. Two grams of paper were cut into small pieces, weighed in and added to ASE vials. The prepared samples were then spiked with the spiking solution to obtain 0.1, 0.2 and 0.5 ng of each PFAS per gram of paper. These concentrations were selected because they reflected the expected concentrations of PFAS in the samples. Finally, to each sample 100 µL of internal standard solution in concentration 0.01 ng mL^{-1} was added, yielding a total amount of each internal standard of 1 ng.

Combustion ion chromatography

For quantification, a calibration curve from 0.2 to 10 mg L^{-1} in ultrapure water (Milli-Pore, 18 M Ω

cm) was prepared using single element fluoride standard (ROTI Star, 1000 mg L^{-1} F^- in water). 100 µL was injected into a pre-baked ceramic sample boat containing glass wool for better dispersion of the fluids. Ultrapure water was used as the calibration blank, and an empty ceramic boat was used as the method blank. The calibration standards and blanks underwent the same combustion and ion chromatographic program as the samples. The limits of detection (LOD) and quantification (LOQ) were calculated by 3 times and 10 times the standard deviation of the blank divided by the slope of the calibration curve, respectively.

To assess the instrument recovery, we analysed a certified reference material (BCR®-461, fluorine in clay). Additionally, 10 mg of one virgin fibre paper sample, which contained no detectable PFAS, was spiked with 100 µL of 5 mg L^{-1} calibration standard. To demonstrate that the instrument is producing reliable results, independent of the solvent composition, we analysed ~1 mg F L^{-1} of PFOA in methanol/water (1:1, *v/v)* and in methanol.

Targeted analysis using LC-MS/MS

The samples are analysed for the presence of 23 individual PFAS using liquid chromatography coupled with a triple quadrupole mass spectrometer (Shimadzu 8050, Shimadzu Corporation, Japan) equipped with a 50 × 2.1 mm, 2.7 µm particle size Restek Raptor C18 column and 50 × 2.1 mm, 5 µm particle size Restek PFAS delay column (Restek, Pennsylvania, USA).

Mobile phase A consisted of 5 mM ammonium acetate in water, and mobile phase B was methanol. A gradient profile started at 20% B and continued with a linear range to 95% B up to 8 min when it was ramped again to 20% B and held for two more minutes to allow the column to reequilibrate. The total acquisition time was 10 min. The separation was performed at 40 °C; the flow rate and injection volume were set at 0.4 mL min^{-1} and 5 µL, respectively.

Qualitative and quantitative analyses were made using the multiple reaction monitoring (MRM) method in negative mode. Optimized identification of analysed compounds was achieved with an interface voltage of 0.5 kV, nebulizing gas flow at 2 L min^{-1}, heating and drying gas flows set at 10 L min^{-1}, with interface temperature at 300 °C. Detailed information on instrumental parameters, and diagnostic transitions is provided in Supplementing material (Table S2).

Instrumental operations, data acquisition and data analysis were performed with the Lab Solutions software (version 5.114, Shimadzu Corporation).

Total fluorine and extractable organic fluorine analysis using combustion ion chromatography (CIC)

Total fluorine analysis is carried out using a Thermo-Mitsubishi Analytech CIC as described previously (Schröder et al. 2024). Briefly, for total fluorine analysis, milled neat solids, approximately 10 mg each, were placed in pre-baked ceramic sample boats. For extractable organic fluorine, 100 µL of extracts in methanol without internal standard, prepared as described previously, were used. Samples were introduced into a combustion oven (HF-210, Mitsubishi Analytech) and heated to 1100 °C with argon (200 mL min^{-1}), argon with water (100 mL min^{-1}) and oxygen (400 mL min^{-1}). All combustion gases were collected in ultrapure water (GA-210, Mitsubishi Analytech), and the absorption solution (ultrapure water) was adjusted to 10 mL. An aliquot of 100 µL was injected onto a Dionex IonPac AS20 column (2 × 250 mm) equipped with a guard column (Dionex IonPac AG19 (2 × 50 mm)). Chromatographic separation was performed using a gradient elution of potassium hydroxide (8 – 60 mM L^{-1}).

Calculation of migration to food matrices

To avoid the laborious and time-consuming process of performing migration experiments for fifty-seven samples, we calculated the theoretical migration into the food of PFAS detected in the samples. The calculation of migration was based on the definition of the European Commission on plastic materials and articles intended to come into contact with food, which assumes that 1 kg of food is in contact with 6 dm^2 of packaging material (European Commission 2011). Migration was

calculated based on the results of targeted analysis for all the paper samples in which PFAS were detected above LOQ, from the total concentration of PFAS and the mass of the paper samples, assuming that all of the analytes would migrate into food, representing the 'worst-case' scenario.

Results and discussion

Quality control

LC-MS/MS

Seven calibration solutions of native PFAS in concentrations ranging from 0.01 to 5 ng mL^{-1}, with the labelled standard in concentration of 1 ng mL^{-1}, were measured in triplicates, as described previously. From the results, LOQs for each analyte as well as linearity were obtained. LOQs were estimated as the lowest concentration for which the peak area was at least ten times signal to noise ratio, and expressed in ng g^{-1} of paper. For obtaining the recovery of each analyte, paper samples were spiked with a native PFAS solution to yield 0.1, 0.2 and 0.5 ng g^{-1} and labelled PFAS solution in a concentration of 0.5 ng g^{-1}, in duplicates, and extracted and analysed as described above. Results are presented in Supplementary material (Table S3).

LOQs for individual analytes were in the range between 0.025 and 0.1 ng g^{-1}. Linearity for each analyte was above 0.99, showing that the method is robust and yields accurate results. The method displayed good overall recovery, with average values at 70 ± 13, 78 ± 15 and 90 ± 26%, at spiking concentrations of 0.1, 0.2 and 0.5 ng g^{-1}, respectively.

Combustion ion chromatography

For the total extractable organic fluorine, the method LOD and LOQ were 30 and 90 ng g^{-1}, respectively. For the reference material BCR-461, a recovery of 92% was obtained. Recoveries of PFOA in methanol and methanol: water (1:1, *v/v*) were 96 and 115%, while a virgin paper sample spiked with the 5 mg L^{-1} calibration point had a recovery of 78%. The average relative standard deviation between the triplicates was 5%. The calibration curve of elemental fluoride yielded a linearity of 0.9997, proving that the method is robust and produces accurate results.

Total fluorine and extractable organic fluorine content

A subset of 20 samples, selected from each category of materials, was analysed for total fluorine and extractable organic fluorine. The total fluorine content in the samples ranged from 7×10^3 to 131×10^3 ng g^{-1}. Extractable organic fluorine was under the LOD in all the samples (30 ng g^{-1}). Results from TF and EOF analysis, as well as their comparison with targeted analysis results, can be found in Table 2. Methanol extracts a broader range of compounds than those targeted in our analysis. This includes PFAS that are not amenable to ionization via electrospray, such as certain fluorinated alkanes/alkenes and PFAS precursors, as well as other methanol-soluble PFAS not included in our target list. Consequently, the presence of EOF content is expected to exceed the combined levels of target PFAS. However, all extracts exhibited EOF content below the LOD. Total fluorine content includes inorganic fluoride, and could also include some organofluorines that are less or not extractable in methanol, like polymeric PFAS or lipophilic species, such as fluorinated alkanes. Given that the targeted analysis measures only a small fraction of the PFAS in the sample, it was assumed that the EOF method could indicate how much of the total amount of PFAS in the sample is missed. However, because of relatively high LOD and LOQ for EOF, this method did not provide sufficient information to make this estimation. Nevertheless, it is important to emphasize the difference between the content of total fluorine and extractable organic fluorine. Total fluorine content is measured in the μg g^{-1} range, meaning it is at least three orders of magnitude higher than the organic fluorine. Although all of the measured samples had EOF under the LOD, targeted analysis did reveal certain fluorinated species. Therefore, it would be safe to assume that even the trace amount of inorganic fluorine, that could potentially end up in the extract, would still impact the results of EOF to a significant extent.

Table 2. Comparison of total fluorine content, extractable organic fluorine and results of targeted analysis expressed as the sum of all detected PFAS (∑PFAS [ng g^{-1}]), and calculated for fluorine ([ng F g^{-1}]).

	Sample type	Total fluorine analysis [ng F g^{-1}]	EOF [ng F g^{-1}]	Targeted analysis ∑PFAS [ng g^{-1}]	Targeted analysis [ng F g^{-1}]
Sample 1	Virgin fibre paper	119×10^3	<LOD[a]	<LOD[b]	<LOD
Sample 2	Virgin fibre paper	31×10^3	<LOD	<LOD	<LOD
Sample 3	Fresh fibre pulp	56×10^3	<LOD	<LOD	<LOD
Sample 4	Fresh fibre pulp	9×10^3	<LOD	<LOD	<LOD
Sample 5	Virgin fibre paper	63×10^3	<LOD	<LOD	<LOD
Sample 6	Virgin fibre paper	131×10^3	<LOD	<LOD	<LOD
Sample 7	Fresh fibre pulp	31×10^3	<LOD	<LOD	<LOD
Sample 8	Virgin fibre paper	39×10^3	<LOD	<LOD	<LOD
Sample 9	Secondary paper	29×10^3	<LOD	0.08	0.05
Sample 10	Virgin fibre paper	47×10^3	<LOD	<LOD	<LOD
Sample 11	Virgin fibre paper	46×10^3	<LOD	<LOD	<LOD
Sample 12	Virgin fibre paper	18×10^3	<LOD	<LOD	<LOD
Sample 13	Fresh fibre pulp	7×10^3	<LOD	<LOD	<LOD
Sample 14	Secondary paper	36×10^3	<LOD	0.19	0.12
Sample 15	Secondary paper	57×10^3	<LOD	0.45	0.33
Sample 16	Virgin fibre paper	92×10^3	<LOD	<LOD	<LOD
Sample 17	Virgin paper	47×10^3	<LOD	<LOD	<LOD
Sample 18	Secondary pulp	63×10^3	<LOD	0.29	0.19
Sample 19	Secondary paper	66×10^3	<LOD	0.13	0.09
Sample_20	Secondary paper	84×10^3	<LOD	0.15	0.09

[a]LOD (EOF) – 30 ng g^{-1}
[b]LOD (targeted analysis) – see Table S3.

Comparing the results from TF analysis of different types of samples, it is noticeable that fresh fibre pulp samples have a significantly lower amount of total fluorine (7 – 56×10^3 ng F g^{-1}) than virgin fibre paper (31 – 131×10^3 ng F g^{-1}) or secondary paper and pulp (29 – 84×10^3 ng F g^{-1} and 63×10^3 ng F g^{-1}, respectively). There was no noticeable correlation between the amount of TF and calculated amount of fluorine in targeted PFAS (Supplementary Information, Figures S1 and S2).

From the results for TF and targeted analysis, it was calculated that sum of target PFAS amount to an average of 2.7×10^{-4}% of total fluorine content. These results demonstrate that there is a big gap between the amount of fluorine compounds that can be detected with LC-MS/MS and CIC.

Targeted PFAS analysis

For the targeted analysis, 57 samples were analysed for the presence of 23 PFAS. Among these samples, 33 were classified as secondary papers, two as secondary pulp, 16 as pure virgin fibre papers, and six as fresh fibre pulp. None of the targeted analytes was detected in any of the fresh fibre pulp or virgin paper samples. However, in all of the secondary paper and pulp samples, some of the analytes were detected. In 12 out of the 35 samples, all the analytes that were above the LOD were below the LOQ (Table S3). Nevertheless, it is significant that among those samples, the most frequently detected analytes were PFOS in 75% of the samples, perfluorohexanoic acid (PFHxA), PFOA, perfluoroundecanoic acid (PFUdA), and perfluorododecanoic acid (PFDoA) in 67% of the samples, perfluorodecanoic acid (PFDA) in 58%, perfluorotetradecanoic acid (PFTeDA) in 42%, perfluorotridecanoic acid (PFTrDA) in 25% of the samples, whereas perfluorobutane sulfonate (PFBS), 1H,1H,2H,2H-perfluorohexane sulfonate (4:2FTS), 1H,1H,2H,2H-perfluorooctane sulfonate (6:2FTS), N-methyl-perfluoroctanesulfonamidoacetic acid (N-MeFOSAA) and N-ethyl-perfluoroctanesulfonamidoacetic acid (N-EtFOSAA) were detected in one sample each. The composition of samples that were analysed using targeted analysis, as well as the portion of samples in which PFAS were quantified, are represented in Figure 1.

In the samples where the concentration of PFAS was above the LOQ, PFOA and PFOS were the most prevalent substances, with PFOA present in 90% of the samples, averaging 0.04 ng g^{-1}, and PFOS found in 62% of the samples, averaging 0.07 ng g^{-1}. Additionally, PFHxA was detected in 33% of the samples, with an average concentration of 0.02 ng g^{-1}, while PFBS was identified in 29% of the samples, with an average value of 0.15 ng g^{-1}. PFTrDA was observed in 14% of the samples, with an average concentration of 0.04 ng

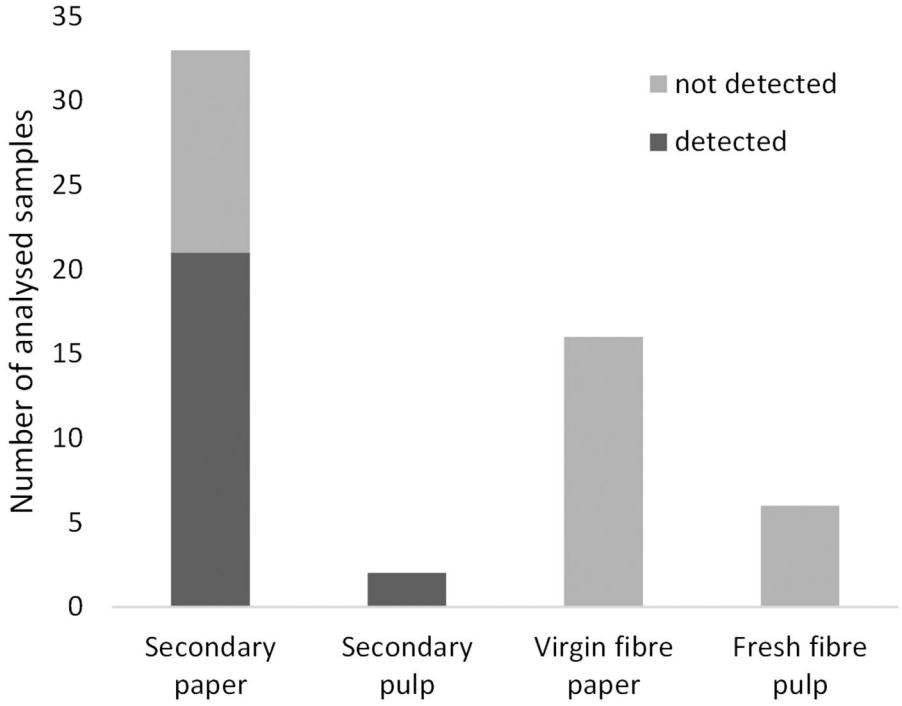


Figure 1. Composition of samples analysed via LC-MS/MS with the portion of samples in which PFAS were quantified.

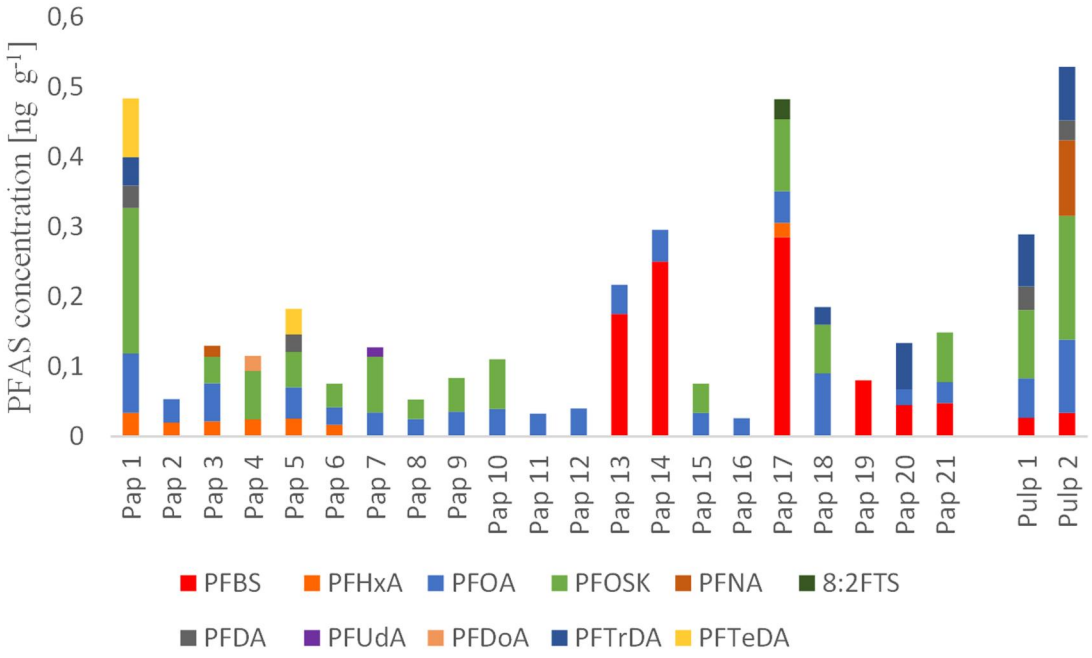


Figure 2. Sums of individual PFAS concentrations detected in samples in which analytes were above LOQ.

g^{-1}. PFTeDA and PFDA were quantified in two samples each, with average values of 0.06 and 0.02 ng g^{-1}, respectively. Lastly, PFUdA and 1H,1H,2H,2H-perfluorodecane sulfonate (8:2FTS) were found in one sample each, at 0.01 and 0.03 ng g^{-1}, respectively (Figure 2).

In previous studies on the content of PFAS in FCM, PFAS were present in considerably higher concentrations. For instance, microwave popcorn bags were reported to have higher PFAS content, ranging up to 500 ng g^{-1} (Begley et al. 2005; Martínez-Moral and Tena 2012; Gebbink et al. 2013; Zabaleta et al. 2017). Whereas, in other FCMs, like pizza boxes, muffin cups or coated paper, PFAS were detected in concentrations up to 40 ng g^{-1} (Moreta and Tena 2013; Granby and

Table 3. Calculated migration of PFAS based on targeted analysis results – assumption of 100% migration.

	Paper grammage [g m^{-2}]	PFAS quantified using LC-MS/MS[ng kg^{-1}]	Modelled concentration of PFAS in food [ng kg^{-1}]
Pap 1	186	480	5.40
Pap 2	125	50	0.40
Pap 3	170	130	1.32
Pap 4	160	120	1.11
Pap 5	135	180	1.48
Pap 6	135	80	0.61
Pap 7	135	130	1.03
Pap 8	125	50	0.39
Pap 9	170	80	0.85
Pap 10	100	110	0.66
Pap 11	125	30	0.24
Pap 12	135	40	0.32
Pap 13	135	220	1.76
Pap 14	186	300	3.30
Pap 15	135	80	0.61
Pap 16	120	30	0.19
Pap 17	170	480	4.93
Pap 18	135	190	1.50
Pap 19	90	80	0.43
Pap 20	130	130	1.04

Tesdal Håland 2018; Sapozhnikova et al. 2023). Our study measured up to 0.53 ng g^{-1} as the sum of target PFAS, which is orders of magnitude lower than the concentrations reported in the above-mentioned FCM. As the samples from our study are used for further processing into FCM, it is possible that the much higher concentrations of PFAS in previous studies are a result of the further processing steps. However, more research is needed to be done in order to validate this hypothesis.

Estimation of the concentration of PFAS in food – migration from the packaging materials

The concentration of PFAS in food matrices was calculated using results from targeted analysis, under the assumption of 100% migration, representing the 'worst-case' scenario. Depending on the mass of the paper sample and based on the definition of 1 kg of food being in contact with 6 dm^2 packaging material, the maximum migration was calculated. The results are displayed in Table 3.

Values obtained for the 'worst-case' migration vary by more than one order of magnitude and were in a range between 0.24 and 5.40 ng kg^{-1} of food. To go one step further, the uptake of PFAS was calculated using the assumption of the European Commission that an average person with a body weight of 60 kg consumes 1 kg of food per day packed in this material (The Council of Europe 2009). The potential uptake of PFAS ranged from 0.01 to 0.09 ng kg^{-1} of body weight per day. Even when taking into account all quantified PFAS, the calculated values show that none of the samples would exceed the EFSA's recommendation for the limit of the combined exposure to four PFAS in food to 4.4 ng kg^{-1} body weight per week. Nevertheless, it is important to acknowledge that FCM might contain additional PFAS that are capable of migrating, yet escaping detection through routine PFAS analysis (LC-MS/MS). This could be due to their inability to ionize with electrospray ionization or because they are not included in the targeted list.

Although there is limited information on the migration behaviour of PFAS from paper-based FCM to food, it would be safe to say that this calculation represents a large overestimation of migration, as previous research has shown that only 0.2–11% of perfluoro carboxylic acids migrate to foodstuff (Elizalde et al. 2018; Zabaleta et al. 2020). However, to determine the actual migration of PFAS from packaging material to foodstuff, it would be necessary to carry out migration tests in food and food simulants.

Conclusion

Results from the targeted analysis show that the samples originating from different phases of the

production process of food packaging materials have a relatively low content of PFAS, ranging from < LOQ to 0.53 ng g^{-1}. Comparing the results from fresh fibre paper and pulp versus secondary products, it can be assumed that the primary contamination with PFAS happens during the recycling process. Most frequently found PFAS belong to the group of legacy PFAS, i.e. perfluorocarboxylic acids with longer carbon chains (C6–C14). As expected, the most present PFAS were PFOA and PFOS, in average concentrations at 0.04 and 0.07 ng g^{-1}, respectively. PFBS exhibited the highest concentration compared to other analytes, with an average value of 0.15 ng g^{-1}. This finding confirms the general trend in recent years in replacing legacy PFAS with shorter-chain analogues (Wang et al. 2013; Ruan et al. 2017; Glenn et al. 2021; European Environment Agency 2023).

Targeted analysis offers only limited insight into the total amount of organic fluorinated compounds that could be present in the samples. It is known from previous research that there is a large gap between the amount of fluorine compounds detected using LC-MS/MS and the amount that is actually present in the sample (Fiedler et al. 2021). Additionally, our results show that the total amount of fluorine in the sample does not correlate with the amount of specific PFAS that were detected and quantified. Therefore, for reliable and more accurate results, it would be necessary to employ several analytical techniques, which would complement each other and give more information about the content and composition profile of PFAS in the samples.

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